S^{2}DMSC: A Self-Supervised Deep Multilevel Subspace Clustering Approach for Large Hyperspectral Images

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***Abstract*— Subspace clustering (SC) has achieved remarkable success in hyperspectral images (HSIs) due to the powerful representation ability of handling high-dimensional complex data. However, most of the existing SC methods focus on linear subspace representation and ignore the more effective nonlinear representation. Besides, SC suffers from the bottlenecks, such as high computation load and memory capacity, due to the spectral decomposition of adjacency matrix for large HSIs. To overcome these limitations, we propose an end-to-end learnable network framework for large HSIs, called self-supervised deep multilevel SC (S^{2}DMSC), which incorporates the convolutional neural network (CNN) module, MSC module, and high-quality pseudo-label-based self-supervised learning module into a unified learning framework. More concretely, the deep multilevel spatial–spectral representation from hierarchical superpixels is modeled as a sparsity-constrained self-expression module for SC to construct high-quality pseudo-labels to learn the network parameters and produce better clusters for hyperspectral pixels. Experimental results on four classical HSIs demonstrate the effectiveness of S^{2}DMSC and exhibit superior clustering performance compared with the representative clustering methods.**

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***Index Terms*— Hierarchical superpixels, high-quality pseudo-labels, multilevel subspace clustering (MSC), self-supervised learning.**

I. Introduction

HYPERSPECTRAL remote sensing technology utilizes imaging spectrometers to obtain hyperspectral images (HSIs) with spatial information, radiation information, and spectral information and is widely used in many application fields, such as mineral resources exploration [1], ocean ecosystem protection and security [2], crop specification and monitoring [3], ecological environment monitoring [4], and medical diagnosis and surgery [5]. In these applications, HSIs clustering is a fundamental technique to analyze and process remote sensing images. The aim of HSIs clustering is to assign unlabeled hyperspectral pixels into multiple nonoverlapping clusters, in which the pixels within a cluster are as similar as possible, and the pixels between clusters are as different as possible. As we all know, HSIs are characterized by high dimensionality, large spectral variability, and complex spatial and spectral structures, which makes clustering to become a challenging task in HSIs processing.

In the past few years, subspace clustering (SC) has been proven to be powerful in HSIs compared with clustering algorithms that rely on proximity measures between hyperspectral pixels [6]. As is known to all, the high-dimensional hyperspectral dataset can be modeled as pixels drawn from the union of multiple low-dimensional subspaces. Early SC relied on the self-representation attribute; in other terms, each pixel in the subspace can be effectively expressed by a linear combination of other pixels in the subspace [7]. Common SC first utilizes sparse or low-rank-constrained self-representation features to create a more accurate affinity matrix and then uses spectral clustering to achieve the clusters of segmentation. Typical SC includes sparse SC (SSC) [8], spectral curvature clustering (SCC) [9], low-rank SC (LRSC) [10], collaborative SC (CSC) [11], scalable SSC (SSSC) [12], sketched SC [13], and so on.

The SC introduced above relies on the linear assumption of self-expression. However, in practical applications, the information of HSIs expressed by the linear SC is very limited when exploring the nonlinear structure features of the HSIs. Recently, deep learning has demonstrated exceptional

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ability to capture the discriminative feature and complex underlying structure of HSIs [14], [15]. Furthermore, the deep neural network model is introduced into the SC algorithm to excavate deep spatial–spectral structure features for HSIs clustering [16], [17]. Most of the early deep SC (DSC) builds on end-to-end convolutional autoencoders with minimal reconstruction errors. For instance, Ji et al. [18] present a DSC network building upon deep autoencoders and a fully connected self-expression layer, which maps the input data into a latent space to construct the affinity matrix between all data. In [19], the convolutional autoencoders and low-rank-constrained self-expression layer are incorporated into an end-to-end network architecture to generate the low-rank representations of favorable SC. In [16], deep spatial–spectral SC builds on the multiscale autoencoders and collaborative-constrained self-expression layer to excavate the spatial–spectral structure information of HSIs. The autoencoder-based DSC is comprised of a reconstruction module and a self-expression module, which neglects the clustering quality. In addition, for the autoencoder-based DSC, feature learning and spectral clustering are applied in two separate stages, lacking effective supervision information to learn deep features.

As a branch of unsupervised learning, self-supervised learning establishes a new research line linking deep learning and SC to develop DSC. The key point of self-supervised learning is to generate supervision information of the data itself, which guides the deep learning model to capture the underlying patterns of the data. Inspired by this, the self-supervised DSC (S^2DSC) has been successfully applied in the field of DSC. Specifically, in [20], self-supervised convolutional SC is proposed to combine the feature representation module, self-expression module, and spectral clustering module into an end-to-end trainable framework. In [17], a trainable S^2DSC is utilized to supervise the affinity matrix learning by the high confidence pseudo-labels. Sun et al. [21] proposed self-supervised deep multilevel SC (S^2DMSC), in which the clustering results from spectral clustering supervise the latent representation learning for each view and common latent subspace learning. In the S^2DSC, the SC is integrated into the deep neural networks to generate the pseudo-labels supervising deep feature learning. However, we can observe that the clustering performance of S^2DSC heavily depends on deep representation features. Therefore, it is necessary to learn the spatial and spectral features of HSIs that are valuable for clustering performance. In addition, for the SC module, it is necessary to calculate the self-representative adjacency matrix and perform singular value decomposition (SVD) for the Laplacian matrix of adjacency matrix. Therefore, the SC module in the S^2DSC framework encounters the bottlenecks, including out-of-memory capacity of GPU or CPU devices and high computational load, for large HSIs.

To deal with the fore-mentioned challenges in S^2DSC, in this article, we focus on self-supervised learning in unsupervised clustering and proposed S^2DMSC for large HSIs. As shown in Fig. 1, the learnable network framework proposed in this article mainly includes three modules: convolutional neural network (CNN) module, MSC module, and high-quality pseudo-label-based self-supervised learning module. The proposed S^2DMSC method incorporates the MSC module into the CNN module, which models the multilevel features from multiple convolutional layers as sparse-constrained self-expression features to construct pseudo-label-based supervision information. Specifically, the MSC module from hierarchical superpixels is proposed to generate pseudo-labels for pixels, which can overcome the computational and storage limitations of CPU and GPU devices for large HSIs. More specifically, the layer-specific deep features in CNN are used to guide region merging between superpixels and form deep multilevel superpixel features; then, graph convolution (GConv) operation is incorporated into the deep multilevel superpixel features to model as a sparse-constrained self-expression matrix for SC, which aggregates spatial neighborhood information and can be more suitable for SC for HSIs. In order to make the CNN module more accurate and produce better deep representations for the adjacency matrix of SC, high-quality pseudo-label-based cross-entropy loss is further proposed to alternately update network parameters and produce better nonoverlapping clusters for hyperspectral pixels.

In summary, we highlight the main contributions of this article as follows.

1) An end-to-end learnable S^2DMSC framework is proposed to effectively take the advantages of CNN and SSC, in which the high-quality pseudo-label supervision module is utilized to update the CNN module and SSC module.
2) We propose a deep MSC module from hierarchical superpixel segmentation, which combines low-level and high-level semantic information of HSIs to learn multilevel representation features beneficial to SC. In particular, the layer-specific features in CNN guide the region merging for the level-specific adjacent superpixels to build hierarchical segmentation with fine-to-coarse; then, we add multiple paths concatenating each hierarchical segmentation to construct the multilevel superpixel features. Besides, the GConv operation is incorporated into the multilevel representation features to aggregate the spatial structure features of HSIs.
3) In order to cope with the challenges of out-of-memory capacity and high computational load for SSC in large HSIs, we propose an MSC module to generate high-quality pseudo-label supervision information. The adjacency matrix of hierarchical superpixels is calculated by concatenating superpixel features of multiple convolutional layers into a sparse-constrained self-expression module, which makes the SSC more robust.

II. Related Works

A. Subspace Clustering

The SC divides the original feature space of hyperspectral pixels into multiple low-dimensional subspaces. For the last decade, SC has attracted increasing attention in unsupervised learning of HSIs, and many SC algorithms have been successfully advanced. Among them, SSC [8] and LRSC [10] are the most classic and bare-bones SC methods in high-dimensional HSIs. Furthermore, the spatial and spectral

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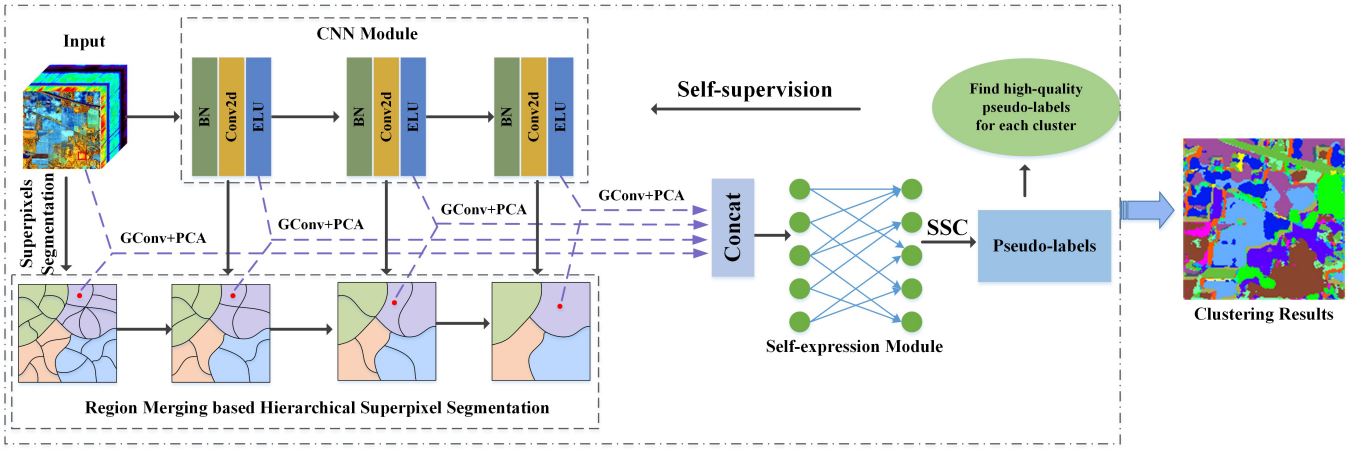

Fig. 1. Framework of the proposed S²DMSC method.

features of HSIs were integrated into the SC to explore the spatial contexture information and spectral similarity. Specifically, Zhang et al. [22] incorporated spectral-weighted spectral constraint and spatial neighborhood information into the SSC framework to build a more accurate similarity matrix between hyperspectral pixels. Huang et al. [23] integrated manifold-based hybrid hypergraph into the unified multiview SSC framework to merge the local and nonlocal spatial contextures of HSIs. Zhang et al. [24] proposed a polygon structure-guided HSI classification method, which gives new thoughts and directions in small sample even one sample. Moreover, to overcome the high memory capacity and computational load for sparse recovery-based SSC, Peng et al. [12] proposed SSSC for large-scale data, which is an out-of-sample extend of SSC.

Recently, many works concentrate on the nonlinear SC algorithms to explore the nonlinear structure features of HSIs, and it is mainly described from two aspects: kernel-based SC [25], [26] and deep learning-based SC [27], [28]. Kernel-based SC transforms the original feature space of nonlinearly structured data to a high-dimensional separable feature space using the kernel function with mercer condition. Common methods include kernel SSC [25], low-rank kernel SC [26], and multiple kernel SC [29]. The deep learning-based SC learns a set of explicit nonlinear mapping functions from the HSIs to map the pixels to the nonlinear feature space and then utilizes the latent deep feature space to calculate the similarity matrix for SC. Specifically, in [30], a deep extension of SSC with l_1-norm was proposed to connect SC and deep neural networks. Furthermore, the autoencoders combined with a self-expression model were proposed to learn the intrinsic structure representation of data [7], [27], [28]. More recently, self-supervised learning was introduced into the DSC to generate supervision information supervised training of the deep networks model [17], [20], [21], [31], and it has attracted more attention in HSIs clustering.

B. Self-Supervised Learning

As a promising branch of unsupervised learning, self-supervised learning utilizes the pretext tasks designed by the image itself to excavate the supervised information from the unlabeled data itself. Compared with supervised learning, self-supervised learning utilizes the data and pseudo-labels to learn the deep network model, and further can learn the valuable deep representation for deep clustering, where pseudo-labels are automatically generated for pretext task from the input data without involving human-labeled samples. Recently, many self-supervised learning methods have achieved excellent performance in deep clustering. For example, Caron et al. [32] proposed DeepCluster, which utilizes the pseudo-labels generated by the K-means for the embedding representation to update the parameters of CNN. Asano et al. [33] further proposed a self-labeling algorithm by an additional constraint that the pseudo-labels are assigned to equally sized subsets, to overcome all samples assigned to a single cluster for DeepCluster. In [34], deep embedded clustering is proposed to simultaneously learn the embedded representation from the deep neural networks and cluster assignments from the feature space. Lv et al. [35] proposed pseudo-supervised DSC to explore high-level similarities under the guidance of feeding pseudo-labels.

In recent years, contrastive self-supervised learning has attracted much attention in unsupervised feature learning, which aims to group the embedding features of similar samples close to each other and the embedding features of different samples away from each other. Representative contrastive learning methods include MoCo [36], SimCLR [37], MoCo v2 [38], and BYOL [39]. More recently, the contrastive self-supervised learning is incorporated into the deep clustering to assign samples into the clusters according to the discriminative deep features. Li et al. [40] proposed contrastive clustering, which conducts cluster-level and instance-level contrastive learnings by maximizing the similarity of embedding features from positive pairs and minimizing the similarity of embedding features from negative pairs. Zhong et al. [41] combined the representation graph contrastive module with the assignment graph contrastive module to simultaneously learn the discriminative deep features and compact clustering assignments. In [42], the feature model is first trained by the contrastive learning; then, the clustering head is trained by the prototype pseudo-labeling; finally, the reliable pseudo-labeling

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method is utilized to jointly train the clustering head and feature model.

III. Proposed Method

In this section, we proposed S^{2}DMSC to learn clustering-friendly deep features and multilevel subspace representation, and the proposed framework is shown in Fig. 1. First, we introduce the overall network formulation. Then, the deep spatial–spectral multilevel features are incorporated into sparsity-constrained SC module to explore both low-level and high-level semantic information for large HSIs. Finally, the high-quality pseudo-label-based self-supervision information is proposed to update the CNN module and SC module.

A. Network Formulation

As described in previous work, learning clustering-friendly deep representation for HSIs is of great significance to improve the clustering performance of DSC. To capture the local structure features of HSIs, we adopt the CNN stacked by multiple convolutional layers, which extracts the low-level and high-level semantic information from the original HSI.

Assume that $\mathbf{H^{(0)}} = \mathbf{Y} \in R^{H\times W\times D}$ denotes the input for the CNN, where $\mathbf{Y}$ is the input HSI, and D, H, and W are the bands, height, and width of the input HSI, respectively. The CNN consists of L convolutional layers, where the output of the lth ($1 \le l \le L$) layer is $\mathbf{H}^{(l)} \in R^{H\times W\times C^{(l)}}$. The feature learned by the lth layer convolutional operation in CNN is expressed as follows:

$$\mathbf{H}^{(l)} = f\left(\mathbf{W}_i^{(l)} * \widetilde{\mathbf{H}}^{(l-1)} + \mathbf{B}_i^{(l)}\right) \tag{1}$$

where $\widetilde{\mathbf{H}}^{(l)} = BN(\mathbf{H}^{(l)})$ denotes the batch normalization of $\mathbf{H}^{(l)}$, $\mathbf{W}_i^{(l)}$ and $\mathbf{B}_i^{(l)}$ are the convolutional kernel and bias, respectively, $i = 1, \ldots, C^{(l)}$, $C^{(l)}$ denotes the number of the lth layer convolution kernel, $f(\cdot)$ denotes the activation function, and $*$ denotes the convolution operation. CNN extracts low-level and high-level semantic information by stacking multiple convolutional layers to fuse more valuable information for SC from different layers.

B. Deep Multilevel SSC for Large HSIs

For CNN, different convolutional layers exact increasingly complex features of HSIs, which makes it difficult to learn favorable features for SSC from the final outputs of CNN. Besides, the SSC calculates the SVD of the Laplacian matrix of adjacency matrix between hyperspectral pixels, which is computationally and memory constrained by CPU and GPU devices. To solve the above limitations, we propose a novel DMSC, which is mainly divided into two steps: 1) deep feature merging from hierarchical superpixel segmentation and 2) multilevel SSC.

1) Deep Feature Merging From Hierarchical Superpixel Segmentation: Hierarchical superpixel segmentation provides a set of superpixels with different details in which coarse-level superpixels are composed of fine-level superpixels [43]. Let $\{S^{(l)}\}_{l=0}^{L}$ denote the hierarchical superpixels with $L+1$ levels, where $S^{(l)}$ consists of $W^{(l)}$ superpixels, and $W^{(0)} > \cdots > W^{(l)} > \cdots > W^{(L)}$.

It is well known that superpixel segmentation utilizes low-level image properties to partition images into meaningful regions. However, it is difficult to achieve good hierarchical superpixels guided by low-level features of HSIs, and it is too weak to exploit the spatial structure information of HSIs. Therefore, we combine the high-level and low-level features in CNN to construct the hierarchical superpixels. Specifically, the hyperspectral pixels within each superpixel have the characteristics of similar spectral and spatial structures; this means we can take superpixel as the basic unit for hierarchical superpixel segmentation. The lth-level superpixel feature $\mathbf{S}^{(l)}$ can be expressed as follows:

$$\begin{aligned}\mathbf{S}^{(l)} &= \left[\mathbf{h}_1^{(l)}, \ldots, \mathbf{h}_{W^{(l)}}^{(l)}\right] \\ &= \left[\frac{1}{\left|S_1^{(l)}\right|}\sum_{j=1}^{\left|S_1^{(l)}\right|}\mathbf{h}_{1,j}^{(l)}, \ldots, \frac{1}{\left|S_{W^{(l)}}^{(l)}\right|}\sum_{j=1}^{\left|S_{W^{(l)}}^{(l)}\right|}\mathbf{h}_{W^{(l)},j}^{(l)}\right]\end{aligned} \tag{2}$$

where $\mathbf{h}_{i,j}^{(l)}$ denotes the jth pixel in the ith superpixel, $\mathbf{h}_i^{(l)}$ denotes the ith superpixel feature for feature $\mathbf{H}^{(l)}$, $S_i^{(l)}$ denotes the ith superpixel in $S^{(l)}$, $|S_i^{(l)}|$ denotes the number of pixels in the ith superpixel, and $\mathbf{S}^{(l)} \in R^{C^{(l)}\times W^{(l)}}$.

After that, the segmentation for the lth-level superpixels is modeled as an undirected graph partition problem. Assuming that $G^{(l)} = (V^{(l)}, E^{(l)})$ represents the eight-connected undirected graph of the lth-level superpixel feature $\mathbf{S}^{(l)}$, $V^{(l)}$ represents the set of nodes $\mathbf{v}_i^{(l)}$, which is represented by the superpixel feature $\mathbf{h}_i^{(l)}$, and $E^{(l)}$ represents the set of edges connected between nodes. Here, region merging is utilized to partition undirected graph $G^{(l)}$ into several connected subgraphs. Boruvka algorithm [43] and Krushal algorithm [44] are classical region merging methods for segmentation. However, they need to predetermine the similarity measures between nodes. Different from Boruvka and Krushal algorithms, inspired by EdgePool [45], we propose learnable parameter-based edge contraction to construct hierarchical segmentation, in which the learnable parameters are utilized to calculate the score of each edge describing similarity weights between nodes for region merging. Specifically, the score for each edge $E_{ij}^{(l)}$ in the graph $G^{(l)}$ is calculated by a simple linear combination of node features and then iteratively contracts the edge with the high scoring and no new merged node incident; see Fig. 2 for details. For the edge between node i and node j, the score is calculated as follows:

$$\mathbf{r}_{ij}^{(l)}\left(E_{ij}^{(l)}\right) = \text{Aconcat}\left(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}\right) + a \tag{3}$$

where $E_{ij}^{(l)}$ denotes the edge connecting the node $\mathbf{v}_i$ and node $\mathbf{v}_j$, and $A \in R^{1\times 2C^{(l)}}$ and $a \in R$ denote learnable parameters. In particular, higher scores indicate lower similarity between nodes, and lower scores indicate higher similarity between nodes.

To get the final normalized score $\mathbf{z}_{ij}^{(l)}$, a softmax normalization is applied to all connected edges, specifically expressed

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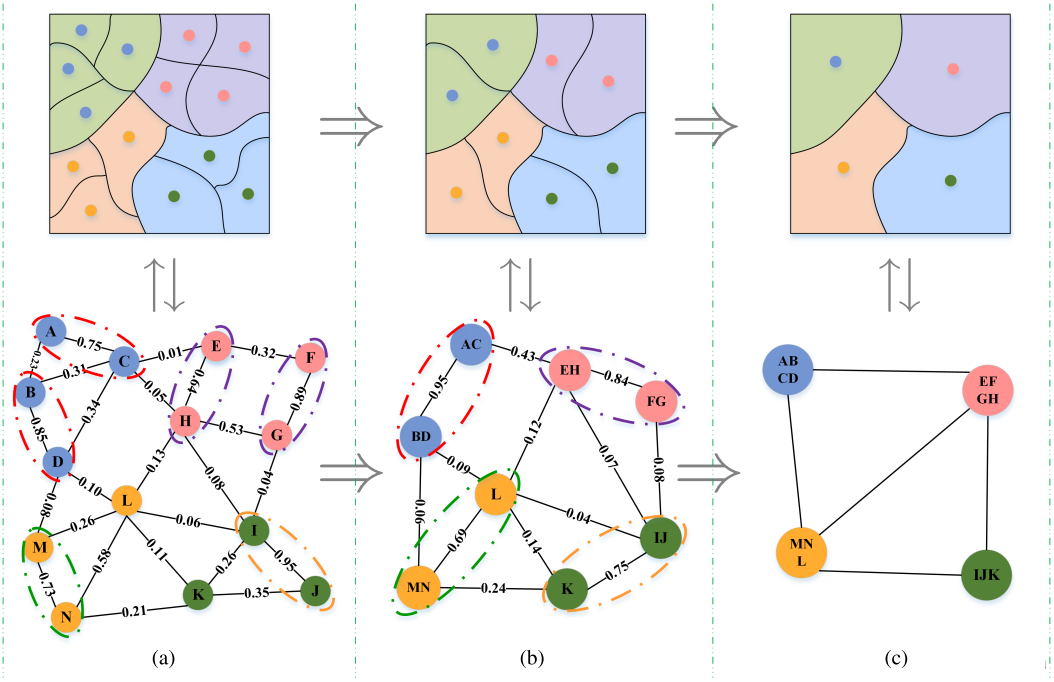

Fig. 2. Hierarchical superpixel segmentation with region merging. In this article, we take superpixel as the basic unit for hierarchical superpixel segmentation. To generate the superpixels $S^{(l)}$ given $S^{(l-1)}$, the $G^{(l-1)} = \left(V^{(l-1)}, E^{(l-1)}\right)$ can be constructed from $S^{(l-1)}$, where the $V^{(l-1)}$ denotes the superpixel features in $l-1$ level, and the $E^{(l-1)}$ is calculated by the learnable parameters. Then, we iteratively contract the edge with the high-scoring and no new merged node incident to obtain the superpixels $S^{(l)}$ (e.g., $\left\{A^{(l-1)}, B^{(l-1)}, C^{(l-1)}, D^{(l-1)}\right\} \rightarrow \left\{(ac)^{(l)}, (BD)^{(l)}\right\}$). By repeating the above procedure multiple times, we obtain the hierarchical superpixels $\left\{S^{(l)}\right\}_{l=1}^{L}$ with L levels from fine to coarse. (a) $(l-1)$th-level segmentation. (b) lth-level segmentation. (c) $(l+1)$th-level segmentation.

as follows:

$$\mathbf{z}_{ij}^{(l)} = \text{Softmax}_{\mathbf{r}_{i:}^{(l)}}\left(\mathbf{r}_{ij}^{(l)}\right). \tag{4}$$

Given scores for all edges, we iteratively contract edges with high scores, ignoring those with newly merged node incident. Repeating the procedure multiple times in this way, we obtain the hierarchical superpixels $\{S^{(l)}\}_{l=0}^{L}$ from fine to coarse. Notably, for the $S^{(0)}$, we utilize the entropy rate segmentation [46] to obtain the initial superpixels replacing the pixels, which can greatly reduce memory capacity and computation load. Moreover, different from Krushal and Boruvka algorithms, the proposed region merging utilizes deep features at different levels to dynamically contract superpixels and further construct the hierarchical superpixels.

By using the progressive region merging for hierarchical superpixels, the lth-level superpixels are mapped into the $(l+1)$th-level superpixels with coarse, and we define the association matrix $\mathbf{Q}^{(l\rightarrow l+1)} \in R^{W^{(l)}\times W^{(l+1)}}$ between $G^{(l)}$ and $G^{(l+1)}$ as follows:

$$\mathbf{Q}_{ij}^{(l\rightarrow l+1)} = \begin{cases} 1, & \text{if } \mathbf{S}_i^{(l)} \subseteq \mathbf{S}_j^{(l+1)} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Here, the association matrix $\mathbf{Q}^{(l\rightarrow l+1)}$ is an invertible matrix. The lth-level superpixels are achieved by simply multiplying the association matrix with the lth layer feature in CNN, as follows:

$$\mathbf{S}^{(l)} = \hat{\mathbf{H}}^{(l)}\mathbf{Q}^{(0)}\mathbf{Q}^{(0\rightarrow 1)}\cdots\mathbf{Q}^{(l-1\rightarrow l)} \tag{6}$$

where $\hat{\mathbf{H}}^{(l)} = \text{reshape}(\mathbf{H}^{(l)}) \in R^{C^{(l)}\times HW}$, $\mathbf{S}^{(l)} \in R^{C^{(l)}\times W^{(l)}}$, and $\mathbf{Q}^{(0)}\mathbf{Q}^{(0\rightarrow 1)}\cdots\mathbf{Q}^{(l-1\rightarrow l)}$ represents the association matrix mapping the hyperspectral pixels to the lth-level superpixels. Meanwhile, the superpixel feature $\mathbf{S}^{(l)}$ is also achieved via averaging inside each pixel in (2); therefore, we can clearly see that (6) is equivalent to (2).

To this end, superpixel features at different levels are obtained by performing the association matrix for the corresponding deep feature $\mathbf{H}^{(l)}$. However, the superpixel features of each level only explore the spatial contexture properties of HSIs and neglect the long-range spatial similarity between superpixels. Therefore, with the purpose of contemporaneously exploiting the long-range spatial relationship and local spatial–spectral features of HSIs, the GConv operation is incorporated into superpixel features of each level to aggregate the neighborhood information of the spatial and spectral, and it can be described as follows:

$$\hat{\mathbf{S}}^{(l)} = \mathbf{S}^{(l)}\left(\mathbf{D}^{(l)}\right)^{-\frac{1}{2}}\bar{\mathbf{W}}\left(\mathbf{D}^{(l)}\right)^{-\frac{1}{2}} \tag{7}$$

where $\bar{\mathbf{W}}^{(l)}$ calculates the adjacency matrix between superpixel features $\mathbf{S}^{(l)}$ by K-nearest neighbor (KNN) [47] algorithm, $\mathbf{D}^{(l)}$ represents the degree matrix of $\bar{\mathbf{W}}^{(l)}$, and $(\mathbf{D}^{(l)})^{-(1/2)}\bar{\mathbf{W}}(\mathbf{D}^{(l)})^{-(1/2)}$ represents the normalization symmetric Laplacian matrix.

2) Multilevel SSC: Since different convolutional layers provide different semantic representations of HSIs, it is very difficult to learn suitable representation features for SC from the final output of CNN. To learn more favorable representations for SC, low-level semantic information is incorporated

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into high-level semantic information to learn more beneficial subspace representation and improve the performance of HSIs clustering.

In general, the number of convolution kernels at different convolutional layers is different. Therefore, we first perform the association matrix $\mathbf{Q}^{(l\to l+1)}$ and principal component analysis (PCA) for the superpixel feature $\hat{\mathbf{S}}^{(l)} \in R^{C^{(l)}\times W^{(l)}}$ to get $\widetilde{\mathbf{S}}^{(l)} \in R^{C^{(L)}\times W^{(L)}}$, i.e., $\widetilde{\mathbf{S}}^{(l)} = \mathrm{PCA}(\hat{\mathbf{S}}^{(l)}\mathbf{Q}^{(l\to l+1)}\cdots\mathbf{Q}^{(L-1\to L)})$. Then, the multilevel semantic representations are incorporated into the SSC to produce well-suited self-expression features for SC, and the commonly used similarity matrix is calculated as follows:

$$\begin{aligned}&\min_{\mathbf{X}}\left\|\bar{S}-\bar{S}\mathbf{X}\right\|_F^2+\lambda\|\mathbf{X}\|_1\\&\text{s.t. } \mathrm{diag}(\mathbf{X})=0\end{aligned}\tag{8}$$

where $\bar{\mathbf{S}} = \mathrm{concat}(\widetilde{\mathbf{S}}^{(L)},\ldots,\widetilde{\mathbf{S}}^{(l)},\ldots,\widetilde{\mathbf{S}}^{(1)})$ represents the concatenation of the all level superpixel features, $\mathbf{X} \in R^{W^{(L)}\times W^{(L)}}$ denotes the sparse representation matrix with diagonal zero constraints to avoid the trivial solution of each superpixel being linearly expressed by itself, and λ denotes the trade-off parameter. The adjacency matrix is calculated by the sparse representation matrix of (8) as follows:

$$\hat{\mathbf{W}} = |\mathbf{X}| + |\mathbf{X}|^T \tag{9}$$

where $\hat{\mathbf{W}}$ is the adjacency matrix of the feature matrix $\bar{\mathbf{S}}$ and $|\mathbf{X}|$ is the nonnegative matrix of X. Then, spectral clustering algorithm [48] is applied for the adjacency matrix $\hat{\mathbf{W}}$ to obtain pseudo-labels for superpixels. Usually, pixels are highly local correlated with each other, and it can be considered that all pixels in the same superpixel belong to the same pseudo-labels. Therefore, the pseudo-labels of superpixels can be mapped into the original image to label all pixels. Specifically, letting $\mathbf{P} \in R^{W^{(L)}\times 1}$ represents the pseudo-label of L-level superpixels, and the pseudo-label mapping from superpixels to all pixels is calculated as follows:

$$\bar{\mathbf{P}} = \mathbf{Q}^{(0)}\mathbf{Q}^{(0\to 1)}\cdots\mathbf{Q}^{(L-1\to L)}\mathbf{P} \tag{10}$$

where $\bar{\mathbf{P}} \in R^{HW\times 1}$. Meanwhile, the inverse mapping from pixels to superpixels is obtained by multiplying the transposition of the association matrix with the pseudo-label of pixels, i.e., $\mathbf{P} = (\mathbf{Q}^{(0)}\mathbf{Q}^{(0\to 1)}\cdots\mathbf{Q}^{(l-1\to l)})^T\bar{\mathbf{P}}$. Finally, the 1-D pseudo-label $\bar{\mathbf{P}} \in R^{HW\times 1}$ is transformed into a 2-D $\bar{\mathbf{P}} \in R^{H\times W}$.

C. Self-Supervision Information for the CNN

In general, CNN requires large amounts of labeled samples to extract deep representation features, which will lead to the failure of CNN directly applied to DSC. In recent years, self-supervised learning relies on pretext tasks with pseudo-label to replace human-labeled samples to train CNN. To this end, we incorporate the self-supervision module into the CNN and self-expression modules to effectively train the network parameters and learn multilevel spatial–spectral representations.

The pseudo-label $\bar{\mathbf{P}}$ is not necessarily assigned correctly clusters corresponding to the ground truth of HSIs, but contains meaningful information about the hyperspectral data [20]. This motivates us to utilize high-quality pseudo-label information to supervise the learning of CNN and self-expression modules. In the pseudo-label-based self-supervised learning module, it is usually desirable to remove incorrect pseudo-labels and find the most reliable pseudo-labels. Therefore, we propose an information entropy-based confidence estimation to measure the uncertainty of the probability labels for CNN. For the hyperspectral pixel $\mathbf{Y}(\mathbf{p}_t)$ at the location $\mathbf{p}_t = (i, j)$, the information entropy is described as follows:

$$I(\mathbf{p}_t) = -\sum_{c=1}^{K}\mathbf{H}_{ijc}^{(L)}\log\left(\mathbf{H}_{ijc}^{(L)}\right) \tag{11}$$

where $\mathbf{H}_{ijc}$ represents the probability of the pixel $\mathbf{Y}(\mathbf{p}_t)$ assigned to the cth cluster and K represents the number of clusters. Next, we set a threshold for the information entropy to select pseudo-labels with high confidence

$$\mathbf{T}_{ij} = \begin{cases}1, & \text{if } I(\mathbf{p}_t) \le \text{thres}\\ 0, & \text{otherwises}\end{cases} \tag{12}$$

where $\mathbf{T}_{ij}$ represents the mask for pseudo-label of the pixel $\mathbf{Y}(\mathbf{p}_t)$ and thres is set as the mean of information entropy for all pixels.

Now, we perform one-hot operation for pseudo-label $\bar{\mathbf{P}}$ to get pseudo-label $\bar{\mathbf{P}} \in R^{H\times W\times K}$. Finally, the probability distribution $\mathbf{H}^{(L)}$ and the high-quality pseudo-label $\bar{\mathbf{P}}$ are utilized to train the CNN module, and the cross-entropy loss function is reformulated as follows:

$$\mathrm{Loss} = -\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{K}\mathbf{T}_{ij}\mathbf{H}_{ijc}^{(L)}\log\bar{\mathbf{P}}_{ijc} \tag{13}$$

where $\bar{\mathbf{P}}_{ijc}$ represents the cth channel of pseudo-label at the location $\mathbf{p}_t$.

Based on the above analysis, the S^{2}DMSC proposed in this article alternately updates the following two stages: 1) perform multilevel SSC for deep spatial–spectral representation to obtain pseudo-labels and 2) high-quality pseudo-labels supervise the CNN and self-expression modules. It is worth noting that the pseudo-label information assigned to a specific pixel is unknown permutations. That is, the pseudo-label information may be inconsistent between two consecutive iterations. To solve this problem, referring to [20], the Huanarian algorithm [49] is applied to find an optimal alignment between pseudo-labels of consecutive iterations. Specifically, the probability feature $\mathbf{H}^{(L)}$ generated by the CNN might not be consistent in consecutive iterations, resulting in mask $\mathbf{T}_{ij}$ inconsistent in consecutive iterations. Therefore, in this article, each updated mask matrix $\mathbf{T}$ is merged into its last update; that is, for the pixel $\mathbf{Y}(\mathbf{p}_t)$, if $\mathbf{T}_{ij} = 1$ in the tth iteration, then $\mathbf{T}_{ij} = 1$ in subsequent iterative updates.

In order to make the proposed S^{2}DMSC more efficient, we apply convolutional autoencoders to pretrain the CNN module and provide the initial self-supervised CNN module. Then, the S^{2}DMSC is performed to fine-tune the CNN module and learn more beneficial multilevel deep features for SC.

IV. Experimental Results and Analysis

We evaluate the effectiveness of S^{2}DMSC using four classical hyperspectral datasets. Representative clustering methods,

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such as K-means [50], SSC [8], USPEC [51], SGCNR [52], SSSC [12], S^{5}C [53], CAE [54], VAE [55], DCN [56], and DEC [34], are utilized for comparative experiments. The benchmark clustering methods in this article fall into centroid-based clustering methods (K-means), subspace-based clustering methods (SSC, USPEC, SGCNR, SSSC, and S^{5}C), and deep learning-based clustering methods (CAE, VAE, DCN, and DEC). In all of the above methods, the number of clusters is set by referencing the ground truth in each dataset. For the K-means and USPEC, the initial parameters selection is very sensitive; thus, we run repeatedly and report the optimal performance. In all clustering methods, the parameters were manually determined to the optimum.

In all experiments, the evaluation metrics perform producer's accuracy (PA), user's accuracy (UA), overall accuracy (OA), homogeneity, completeness, V-measure, accuracy (ACC), Fowlkes–Mallows index (FMI), adjusted rand index (ARI), adjusted mutual information (AMI), and normalized mutual information (NMI) to verify the clustering performances of different methods.

A. Network Setup

S²DMSC consists of three modules: CNN module, MSC module, and high-quality pseudo-label-based self-supervision module. Specifically, the CNN module contains three convolutional layers, the convolution kernel size of the first two layers is set to 5×5, the output channels are set to 128 and 64, respectively, convolution kernel size of the third layer is set to 1×1, and all activation functions are exponential linear unit (ELU). We perform Adam optimizer with a learning rate of $1e-4$ to train the CNN module. In the MSC module, the number of multilevel is consistent with the number of convolutional layers, and the number of neighborhoods in the KNN algorithm is set to 8.

B. Dataset Description and Experimental Analysis

1) Indian Pines Image: AVIRIS spectrometer collected Indian Pines image in the year 1992, which contains 16 kinds of land-cover classes with the size of 145×145. We remove the 20 bands with water absorption and high noise, resulting in 200 ones. Fig. 3 shows the maps that are obtained by different representative clustering methods. Tables I and II report the clustering quantitative evaluation of different clustering mechanisms, each row of which has the bold and underlined lines to indicate the optimal and second-best clustering performances, respectively.

From the clustering accuracies achieved by different clustering methods in Table I, it can be easily observed that S²DMSC achieves the best clustering performances in Indian Pines image. In the subspace-based clustering method, SGCNR utilizes the anchor graph to construct similarity matrix between pixels, and the clustering accuracy OA achieves the best clustering performances. Specifically, compared with SSC, USPEC, SSSC, and S^{5}C, SGCNR improves the clustering accuracy OA by 1.79%, 4.62%, 5.38%, and 2.59%, respectively, especially in the Alfalfa and Oats classes. In the deep learning-based clustering methods, compared with CAE, DCN, and DEC, VAE achieves better clustering results and clustering accuracies by combining normal distribution into the features of hidden layer. Specifically, compared with CAE, DCN, and DEC, VAE achieves better clustering results in grass-pasture class, soybean-notill class, and stone-steel-towers class, which improves the clustering accuracy OA by 4.46%, 9.66%, and 0.38%, respectively. Compared with the centroid-based and deep learning-based clustering methods, subspace-based clustering methods can achieve better clustering performances by constructing the similarity matrix to describe the relationships between pixels. Specifically, compared with K-means and VAE, SGCNR improves clustering performance OA by 4.29% and 2.90%, respectively. Compared with K-means, SGCNR, and VAE, the S²DMSC incorporates high-level and low-level semantic information into the SC to achieve higher clustering accuracy, and the clustering accuracy in terms of OA is improved by 16.10%, 11.81%, and 14.71%, respectively. Furthermore, from the perspective of misclassification of clustering results, we can observe from Table I that the alfalfa class, grass-pasture-mowed class, oats class, and stone-steel-towers class have serious misclassification, because the Indian Pines image has similar and mixed spectral signatures and the problem of imbalanced hyperspectral data, which raise the challenge for the HSIs clustering. Although the S²DMSC has some errors in the alfalfa class and grass-pasture-mowed class, it achieves good classification ability in the oats class and stone-steel-towers class, which further indicates that the proposed S²DMSC can achieve a certain effect on the mixed spectral and unbalanced Indian Pines dataset. In summary, S²DMSC achieves superior clustering performances in both visual map and quantitative accuracy compared with other representative clustering methods, and the best quantitative accuracy in terms of OA is 54.21%.

To further illustrate the effectiveness of S²DMSC, Table II displays the quantitative clustering accuracy with different evaluation metrics for Indian Pines. It can be clearly illustrated from Table II that S²DMSC achieves better clustering performances than other representative clustering methods. From the subspace-based clustering methods, it can find that SGCNR achieves the superior quantitative clustering accuracy. Among deep learning-based clustering methods, VAE obtains the superior clustering accuracy. Compared with K-means, VAE, and SGCNR, S²DMSC gets the best clustering results among clustering evaluation metrics. This further illustrates that the S²DMSC proposed in this article fully exploits the inherent nonlinear structural features of Indian Pines image, improving the clustering performance.

2) Salinas Image: AVIRIS spectrometer collected Salinas image in the year 1998, which contains 16 kinds of land-cover classes with the size of 512×217. The spatial resolution is 3.7 m. We remove the bands with water absorption and high noise (108–112, 154–167, and 20) in the experiment. Fig. 4 shows the maps that are obtained by different representative clustering methods. Tables III and IV report the clustering quantitative evaluation of different clustering mechanisms.

From Table III and Fig. 4, we can clearly observe that the effectiveness and superiority of S²DMSC are remarkable. Subspace-based clustering methods assign clusters for each pixel by constructing similarity graph between pixels; then,

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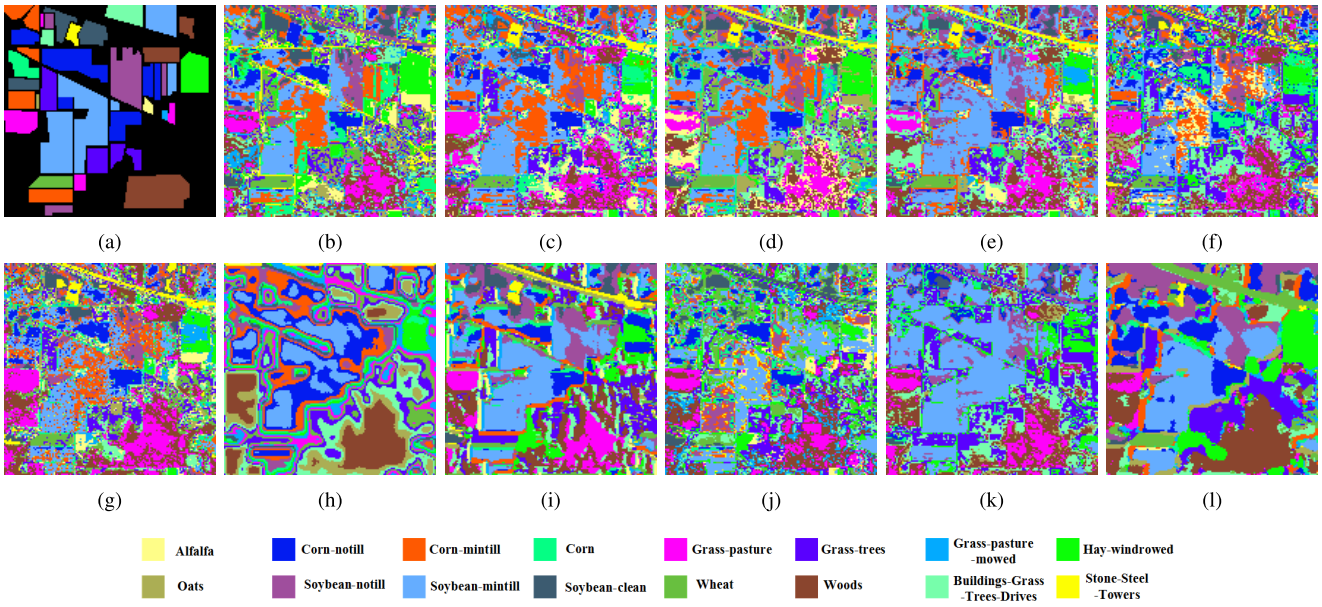

Fig. 3. Clustering maps of Indian Pines image. (a) Ground truth. (b) K-means. (c) SSC. (d) USPEC. (e) SGCNR. (f) SSSC. (g) S^5C. (h) CAE. (i) VAE. (j) DCN. (k) DEC method. (l) S^2DMSC.

TABLE I

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE INDIAN PINES IMAGE [PA, UA, AND OA (%)]

Method	Class	K-means	SSC	USPEC	SGCNR	SSSC	S^5C	CAE	VAE	DCN	DEC	S^2DMSC
Producer's	Alfalfa	11.11	7.41	0	18.52	1.85	11.11	0	0	**74.07**	0	0
accuracy	Corn-notill	26.22	28.73	29.43	30.96	21.97	32.78	31.59	28.38	25.52	6.28	**48.40**
(PA)	Corn-mintill	35.37	**44.36**	40.89	3.12	34.65	32.61	33.69	9.71	8.27	0	8.75
(%)	Corn	**41.45**	2.56	19.23	20.09	32.48	8.97	15.81	22.65	9.40	18.38	11.54
	Grass-pasture	48.69	62.78	39.24	44.87	60.36	**62.78**	14.89	51.91	48.09	45.47	31.59
	Grass-trees	28.92	67.34	30.39	45.25	55.15	59.84	32.26	47.12	33.20	61.58	**83.53**
	Grass-pasture-mowed	0	69.23	0	0	50.00	**76.92**	69.23	11.54	0	0	0
	Hay-windrowed	98.36	58.08	93.25	60.53	53.99	56.03	71.17	41.51	24.54	66.67	**98.98**
	Oats	0	0	55.00	70.00	0	20.00	0	0	0	0	**100**
	Soybean-notill	38.84	21.07	31.92	33.26	27.07	28.51	12.71	45.76	10.64	27.07	**50.62**
	Soybean-mintill	38.01	47.08	40.44	67.46	36.02	41.69	36.75	64.59	37.36	**74.84**	67.99
	Soybean-clean	**29.97**	9.28	18.73	13.19	11.40	18.24	10.91	18.40	18.89	11.73	1.79
	Wheat	90.57	93.40	**95.75**	95.28	77.83	83.49	32.55	26.42	53.30	0.47	93.87
	Woods	37.64	40.42	33.38	42.66	47.76	46.14	72.18	29.29	50.31	48.38	**83.46**
	Buildings-Grass-Trees-Drives	15.79	18.95	20.79	23.95	26.05	8.16	21.05	17.89	21.32	**26.84**	14.74
	Stone-Steel-Towers	0	89.47	87.37	88.42	67.37	83.16	0	**89.47**	3.16	0	27.37
User's	Alfalfa	5.94	1.97	0	14.29	0.12	3.61	0	0	**31.50**	0	0
accuracy	Corn-notill	45.63	53.51	56.27	**56.63**	37.37	49.84	22.58	43.02	50.14	28.39	43.84
(UA)	Corn-mintill	18.19	16.09	18.58	5.37	21.86	15.64	20.06	**35.37**	12.21	0	20.00
(%)	Corn	12.37	1.83	11.60	5.77	10.26	9.21	12.59	11.02	3.91	21.29	**27.27**
	Grass-pasture	34.47	31.11	30.09	31.81	34.68	30.32	22.09	29.97	38.74	30.92	**95.73**
	Grass-trees	69.01	78.23	82.25	81.06	57.54	76.28	54.28	46.44	72.09	40.64	**91.36**
	Grass-pasture-mowed	0	5.10	0	0	**8.02**	7.38	4.84	0.82	0	0	0
	Hay-windrowed	62.39	86.85	**88.72**	87.57	54.21	82.53	73.73	40.60	17.62	41.74	66.85
	Oats	0	0	2.86	**8.00**	0	2.19	0	0	0	0	6.60
	Soybean-notill	**56.54**	39.16	35.44	22.56	34.25	34.72	15.49	29.97	10.72	26.76	28.49
	Soybean-mintill	47.33	46.99	48.90	49.09	52.02	48.47	48.58	55.21	37.71	39.92	**61.56**
	Soybean-clean	25.92	22.89	25.61	27.27	22.29	24.14	**40.85**	26.04	25.66	35.82	15.71
	Wheat	32.76	67.12	33.50	40.81	70.82	49.30	18.60	26.17	12.14	**100**	54.22
	Woods	85.59	88.49	85.21	88.32	84.89	84.08	80.73	59.87	77.78	77.48	**88.74**
	Buildings-Grass-Trees-Drives	20.07	**35.12**	34.05	17.17	23.63	19.62	28.67	19.05	22.50	25.12	30.27
	Stone-Steel-Towers	0	82.52	91.21	82.35	37.87	28.21	0	**97.70**	0.67	0	47.27
OA		38.11	40.61	37.78	42.40	37.02	39.81	35.04	39.50	29.84	39.12	**54.21**

the high-dimensional HSIs clustering is transformed into a graph cut problem. In the subspace-based clustering methods, in comparison with USPEC, SSSC, and S^5C, SGCNR achieves a significant improvement of clustering performance in the lettuce_romaine_7wk class and fallow_rough_plow class; furthermore, the clustering accuracy OA achieves the best clustering accuracies. For the deep learning-based clustering methods, DEC achieves better clustering accuracies. Specifically, in comparison with CAE, VAE, and DCN, DEC improves the clustering accuracy OA by 1.18%, 12.26%, and 9.42%, respectively, by reducing the misclassification of fallow_smooth, lettuce_romaine_6wk, and lettuce_romaine_5wk

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TABLE II

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE INDIAN PINES IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	K-means	SSC	USPEC	SGCNR	SSSC	S^5C	CAE	VAE	DCN	DEC	S^2DMSC
NMI	43.86	45.75	44.70	46.36	39.03	43.05	38.80	39.09	32.39	40.24	**50.31**
AMI	42.17	42.48	42.87	45.83	38.75	42.78	38.51	38.80	32.09	40.01	**50.06**
ARI	21.95	23.18	22.48	26.45	20.24	23.54	21.25	25.15	14.95	22.49	**35.57**
FMI	30.71	33.11	31.28	36.64	28.83	32.40	30.65	34.66	24.61	37.07	**44.36**
V-measure	43.84	42.75	44.67	46.35	39.03	43.05	38.80	39.09	32.39	40.24	**50.31**
Completeness	42.43	42.95	43.12	46.62	37.37	41.93	38.38	38.95	31.34	45.24	**51.93**
Homogeneity	45.35	42.55	46.34	46.09	40.85	44.24	39.23	39.23	33.51	36.24	**48.79**
ACC	38.11	40.61	37.78	42.40	37.02	39.81	35.04	39.50	29.84	39.12	**54.21**

TABLE III

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE SALINAS IMAGE [PA, UA, AND OA (%)]

Method	Class	K-means	SSC	USPEC	SGCNR	SSSC	S^5C	CAE	VAE	DCN	DEC	S^2DMSC
Producer's accuracy (PA) (%)	Brocoli_green_weeds_1	**98.51**	-	0	0	0	0	0	1.00	0	0	0
	Brocoli_green_weeds_2	83.90	-	98.42	96.22	87.98	97.88	96.94	99.03	92.86	85.83	**99.81**
	Fallow	38.01	-	42.00	0.10	0	4.25	81.98	0.05	**85.17**	0	58.35
	Fallow_rough_plow	**99.71**	-	0	99.35	0	83.36	93.69	85.94	90.32	99.50	99.43
	Fallow_smooth	99.22	-	96.71	99.10	**99.89**	76.89	85.88	78.42	78.72	97.39	91.64
	Stubble	99.44	-	99.39	99.39	80.90	**99.47**	97.47	95.45	49.89	96.89	94.06
	Celery	38.53	-	99.36	97.46	0.36	0.03	98.18	88.49	98.58	98.69	**99.47**
	Grapes_untrained	58.72	-	97.18	**98.94**	82.97	52.86	35.52	51.45	91.70	58.58	94.53
	Soil_vinyard_develop	94.44	-	96.07	98.34	96.76	64.24	**98.86**	91.38	87.60	96.84	93.42
	Corn_senesced_green_weeds	56.35	-	59.46	63.79	45.15	24.67	54.97	0.34	0.52	0.03	**66.38**
	Lettuce_romaine_4wk	96.82	-	**98.31**	3.00	0	56.46	0	71.63	74.91	56.09	0.37
	Lettuce_romaine_5wk	0.16	-	45.36	48.26	71.67	67.20	2.02	0	0	99.07	**99.38**
	Lettuce_romaine_6wk	98.03	-	97.49	0	18.45	98.14	81.11	0.11	0	**98.80**	12.34
	Lettuce_romaine_7wk	79.53	-	0	92.90	63.08	89.07	90.84	0.28	0	86.17	**99.81**
	Vinyard_untrained	61.53	-	0	0	18.82	46.62	**79.88**	48.51	0.01	67.83	0.01
	Vinyard_vertical_trellis	0	-	62.48	67.63	72.72	66.52	4.65	3.87	40.56	0	**74.16**
User's accuracy (UA) (%)	Brocoli_green_weeds_1	**99.85**	-	0	0	0	0	0	6.43	0	0	0
	Brocoli_green_weeds_2	56.98	-	64.65	64.02	52.55	38.26	64.81	63.90	63.12	61.43	**64.93**
	Fallow	**97.41**	-	89.25	33.33	0	16.00	59.80	0.21	44.02	0	95.61
	Fallow_rough_plow	98.65	-	0	**99.50**	0	72.90	91.65	23.53	97.30	95.13	94.74
	Fallow_smooth	78.56	-	59.88	68.30	41.61	55.13	94.69	34.31	90.71	78.46	**99.55**
	Stubble	99.52	-	99.95	99.87	100	99.75	100	99.92	100	**100**	99.81
	Celery	48.10	-	86.48	**97.98**	0.73	1.09	88.60	68.51	54.32	81.57	88.62
	Grapes_untrained	73.09	-	58.52	54.78	51.27	**73.89**	64.52	62.81	53.13	70.33	59.77
	Soil_vinyard_develop	97.50	-	87.25	74.96	96.11	70.69	86.86	64.57	63.66	70.40	**99.16**
	Corn_senesced_green_weeds	61.69	-	78.05	89.59	**95.79**	20.11	52.92	47.83	32.08	33.33	58.68
	Lettuce_romaine_4wk	33.70	-	**50.90**	3.62	0	24.17	0	46.48	35.57	50.72	20.00
	Lettuce_romaine_5wk	2.73	-	**84.61**	83.11	45.94	36.94	6.51	0	0	57.67	70.79
	Lettuce_romaine_6wk	35.48	-	48.48	0	**86.22**	82.63	64.33	1.30	0	41.90	13.93
	Lettuce_romaine_7wk	55.84	-	0	66.49	**94.27**	84.64	82.03	1.51	0	52.78	36.06
	Vinyard_untrained	49.94	-	0	0	36.03	46.32	49.40	44.93	1.10	**51.30**	0.59
	Vinyard_vertical_trellis	0	-	51.98	86.30	**99.02**	85.49	4.58	54.69	31.84	0	92.86
OA		67.99	-	69.07	69.41	57.10	55.38	66.13	55.05	57.89	67.31	**72.16**

classes. In comparison with the K-means, the quantitative accuracy of SGCNR in terms of quantitative evaluation OA is improved by 1.42%. In comparison with the DEC, SGCNR improves the clustering accuracy OA by 2.10%. In comparison with K-means, SGCNR, and DEC, S²DMSC improves the clustering performances by reducing misclassification, and its clustering accuracy OA is improved by 4.17%, 2.75%, and 4.85%, respectively. Except for the analysis above, we can also observe that the clustering performances, especially in broccoli_green_weeds_1 class, fallow class, and lettuce_romaine_6wk class, suffer from serious misclassification in Table III. The reason for this result is that the Salinas dataset has mixed spectral and spectral similarity. More importantly, the S²DMSC shows some breakthroughs in the fallow class and lettuce_romaine_6wk class, achieving relatively good clustering performance. In conclusion, S²DMSC gains superior clustering performance in both visual map and quantitative accuracy compared with other representative clustering methods, and the best quantitative accuracy in terms of OA is 72.16%.

In addition, Table IV shows the clustering accuracies of the quantitative evaluation with different working mechanisms. As shown from Table IV, in comparison with other representative clustering methods, S²DMSC achieves superior quantitative clustering performances in all evaluation metrics. It further illustrates the effectiveness and superiority of S²DMSC on Salinas image.

3) Pavia Center Image: ROSIS sensor collected Pavia Center image in Pavia city, Pavia region, Italy, in the year 2003. The Pavia Center image contains a total of 115 spectral bands with a spectral coverage range of 430–860 nm. In this article, we removed noisy bands and some samples with no information. Also, we select a test image with the size of 1096 × 490 and 105 bands, which contains nine different

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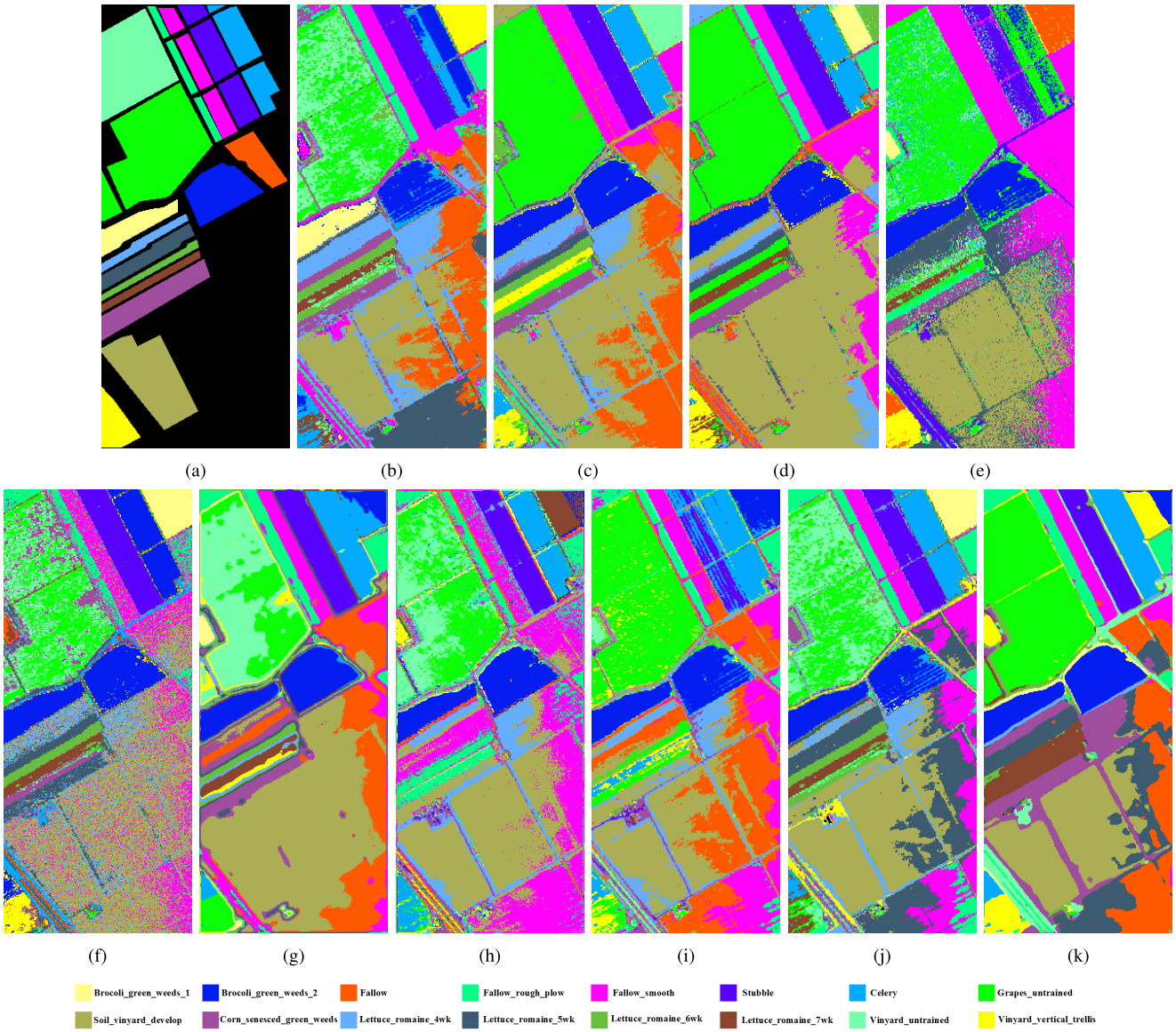

Fig. 4. Clustering maps of Salinas image. (a) Ground truth. (b) K-means. (c) USPEC. (d) SGCNR. (e) SSSC. (f) S^{5}C. (g) CAE. (h) VAE. (i) DCN. (j) DEC. (k) S^{2}DMSC.

TABLE IV

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE SALINAS IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	K-means	SSC	USPEC	SGCNR	SSSC	S^{5}C	CAE	VAE	DCN	DEC	S^{2}DMSC
NMI	73.56	-	78.91	79.86	66.20	64.84	74.12	62.22	66.80	72.94	**82.93**
AMI	72.62	-	72.29	71.57	66.17	64.81	74.10	62.19	66.78	72.92	**82.91**
ARI	54.66	-	63.43	59.22	45.53	43.21	55.41	44.22	49.37	53.85	**66.57**
FMI	59.28	-	70.39	68.06	54.13	49.14	60.17	50.65	58.89	59.07	**72.30**
V-measure	73.56	-	78.61	79.38	66.20	64.84	74.12	62.22	66.80	72.94	**82.93**
Completeness	74.50	-	86.12	**89.08**	72.06	66.28	75.50	66.50	76.02	76.55	88.35
Homogeneity	72.64	-	72.31	71.59	61.22	63.46	72.79	58.46	59.58	69.65	**78.14**
ACC	67.99	-	69.07	69.41	57.10	55.38	66.13	55.05	57.89	67.31	**72.16**

land-cover classes. Fig. 5 shows the maps that are obtained by different representative clustering methods. Tables V and VI report the clustering quantitative evaluation of different clustering mechanisms.

It can be seen from Fig. 5 and Table V that S^{2}DMSC achieves more effective clustering performances and smoother clustering maps on Pavia Center image than other clustering methods. In the subspace-based clustering methods, in comparison with SGCNR, USPEC, and SSSC, S^{5}C improves the clustering accuracy OA by 3.09%, 4.36%, and 2.83%, respectively. In the deep learning-based clustering methods, in comparison with CAE, VAE, and DCN, DEC improves the clustering accuracy OA by 2.26%, 3.84%, and 3.91%, respectively. Compared with K-means, S^{5}C, and DEC, S^{2}DMSC improves the clustering accuracy OA by 13.32%, 8.84%, and 7.70%, respectively. In summary, S^{2}DMSC achieves superior

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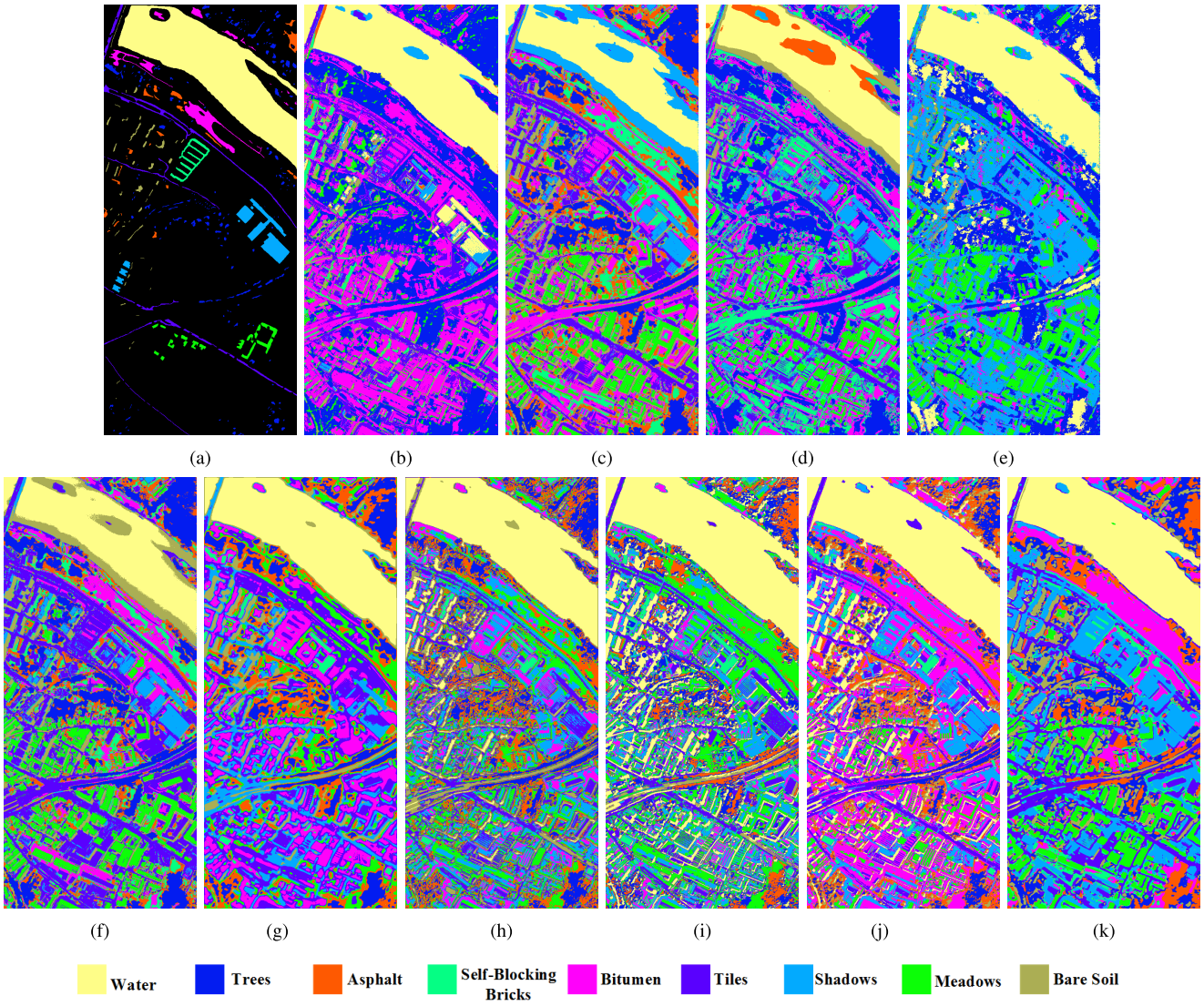

Fig. 5. Clustering maps of Pavia Center image. (a) Ground truth. (b) K-means. (c) USPEC. (d) SGCNR. (e) SSSC. (f) S^{5}C. (g) CAE. (h) VAE. (i) DCN. (j) DEC. (k) S^{2}DMSC.

TABLE V

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE PAVIA CENTER IMAGE [PA, UA, AND OA (%)]

Method	Class	K-means	SSC	USPEC	SGCNR	SSSC	S^{5}C	CAE	VAE	DCN	DEC	S^{2}DMSC
Producer's	Water	97.50	-	77.19	83.23	96.08	85.13	99.23	98.70	99.62	98.93	**99.90**
accuracy	Trees	**99.97**	-	89.33	98.20	65.74	90.01	78.96	70.74	40.65	57.40	54.00
(PA)	Asphalt	0	-	55.03	0	0	50.16	28.36	18.23	86.79	44.21	**88.97**
(%)	Self-Blocking Bricks	0.05	-	0	**77.66**	0	0	0.05	0	0.05	0.05	0
	Bitumen	64.64	-	58.39	46.95	36.01	58.88	50.51	42.89	28.08	44.24	**75.43**
	Tiles	43.77	-	97.89	84.00	11.56	**99.40**	55.47	71.87	87.38	88.73	73.45
	Shadows	17.32	-	83.60	84.51	**99.30**	84.27	43.28	29.53	21.87	71.07	92.51
	Meadows	0	-	98.33	91.96	**99.97**	99.90	0	45.48	45.93	42.66	98.14
	Bare Soil	42.40	-	85.22	22.73	45.40	47.39	90.85	16.44	0.05	0.14	**95.75**
User's	Water	92.85	-	100	100	96.01	**100**	99.79	97.31	96.73	96.75	99.92
accuracy	Trees	65.79	-	78.56	60.49	55.30	79.86	72.94	66.22	92.26	70.21	**98.21**
(UA)	Asphalt	0	-	68.90	0	0	**69.15**	37.93	21.57	39.67	32.13	46.88
(%)	Self-Blocking Bricks	0.20	-	0	**27.43**	0	0	1.23	0	0.18	0.32	0
	Bitumen	36.25	-	82.17	75.48	72.91	**99.95**	40.05	50.63	54.57	49.81	95.05
	Tiles	43.61	-	68.83	86.19	89.51	55.59	35.76	56.22	55.60	72.43	**89.54**
	Shadows	41.03	-	28.97	89.82	35.04	**96.41**	49.03	29.10	26.10	49.01	54.60
	Meadows	0	-	99.97	**100**	69.63	97.93	0	49.00	31.68	93.47	98.84
	Bare Soil	74.57	-	**99.57**	14.41	96.47	9.63	78.30	14.83	0.15	3.49	97.01
OA		77.19	-	77.31	78.58	78.84	81.67	80.55	78.97	78.90	82.81	**90.51**

quantitative clustering precision in terms of quantitative evaluation OA.

Besides, Table VI shows the clustering accuracies of different evaluation metrics on Pavia Center image. As described

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TABLE VI

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE PAVIA CENTER IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	K-means	SSC	USPEC	SGCNR	SSSC	S^5C	CAE	VAE	DCN	DEC	S^2DMSC
NMI	67.53	-	72.43	71.09	64.71	76.06	74.36	71.31	72.85	74.65	**85.24**
AMI	61.96	-	67.85	65.26	64.71	76.05	74.36	71.31	72.85	74.65	**85.24**
ARI	79.14	-	63.51	71.49	79.30	74.48	92.67	88.75	89.37	88.45	**96.03**
FMI	88.49	-	77.32	82.58	88.30	84.49	95.76	93.55	94.01	93.45	**97.70**
V-measure	67.28	-	72.28	70.83	64.71	76.06	74.36	71.31	72.85	74.65	**85.24**
Completeness	73.58	-	67.86	65.27	71.52	72.10	76.94	73.33	75.98	78.53	**86.91**
Homogeneity	61.97	-	77.31	77.44	59.09	80.47	71.95	69.40	69.97	71.14	**83.63**
ACC	77.19	-	77.31	78.58	78.84	81.67	80.55	78.97	78.90	82.81	**90.51**

TABLE VII

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE BOTSWANA IMAGE [PA, UA, AND OA (%)]

Method	Class	K-means	SSC	USPEC	SGCNR	SSSC	S^5C	CAE	VAE	DCN	DEC	S^2DMSC
Producer's accuracy (PA) (%)	Water	68.52	-	100	33.70	91.48	52.59	88.89	92.22	100	98.52	**100**
	Hippo grass	48.51	-	57.43	83.17	67.33	80.20	**83.17**	30.69	0	16.83	0
	FloodPlain grasses 1	0	-	96.02	0	79.28	82.47	39.44	55.78	75.30	**97.21**	0
	FloodPlain grasses 2	70.70	-	22.33	26.51	**88.84**	19.53	65.12	0	0	0	57.67
	Reeds	80.67	-	**83.27**	73.23	66.91	81.04	0.74	32.34	0	80.67	56.88
	Riparian	0	-	0	0	2.60	46.10	11.52	58.36	**60.22**	21.19	49.07
	Firescar	94.21	-	96.91	97.30	81.85	76.45	50.97	44.02	53.28	87.26	**100**
	Island interior	52.71	-	45.32	**68.47**	42.36	43.84	0	66.01	11.33	6.90	0.49
	Acacia woodlands	55.10	-	73.89	70.70	76.43	52.55	28.98	21.97	62.42	63.69	**83.12**
	Acacia shrublands	51.21	-	18.15	0	38.71	20.97	63.31	43.15	62.90	58.47	**97.58**
	Acacia grasslands	**99.02**	-	88.20	95.74	75.41	63.93	34.10	38.69	68.52	55.41	86.56
	Short mopane	72.38	-	76.24	98.34	0	53.59	93.37	27.62	93.92	83.98	**100**
	Mixed mopane	21.64	-	57.84	0	**95.90**	35.82	72.01	0	76.87	56.72	88.81
	Chalcedony	36.84	-	50.53	1.05	84.21	0	83.16	88.42	100	**100**	76.84
User's accuracy (UA) (%)	Water	100	-	100	100	98.41	100	100	100	99.63	98.52	**100**
	Hippo grass	**92.45**	-	82.86	91.30	70.83	78.64	47.46	67.39	0	50.00	0
	FloodPlain grasses 1	0	-	56.97	0	**92.13**	56.71	33.00	22.54	49.35	53.74	0
	FloodPlain grasses 2	75.25	-	28.92	36.08	**90.95**	22.11	27.03	0	0	0	44.93
	Reeds	41.49	-	46.09	53.10	71.15	65.47	3.39	55.41	0	**75.61**	58.40
	Riparian	0	-	0	0	10.77	45.59	34.83	**58.15**	38.57	31.15	44.75
	Firescar	96.44	-	**99.60**	91.30	99.53	60.74	90.41	66.28	99.28	99.56	70.77
	Island interior	71.33	-	**92.00**	50.36	62.32	28.99	0	30.11	56.10	70.00	3.03
	Acacia woodlands	**73.00**	-	69.67	47.44	55.68	65.74	27.91	19.17	35.83	42.11	67.62
	Acacia shrublands	50.00	-	73.77	0	57.14	31.71	70.40	24.37	59.32	47.39	**78.32**
	Acacia grasslands	50.25	-	65.93	36.45	51.45	51.32	45.22	77.12	65.52	50.00	**96.35**
	Short mopane	44.56	-	50.36	33.40	0	47.32	37.89	26.60	60.71	**64.96**	49.32
	Mixed mopane	23.58	-	69.51	0	46.22	49.74	53.46	0	47.58	56.93	**71.90**
	Chalcedony	22.29	-	41.03	**100**	45.98	0	94.05	69.42	62.50	62.09	97.33
OA		54.80	-	63.76	46.58	64.44	52.52	46.83	41.26	55.85	60.16	**67.67**

TABLE VIII

QUANTITATIVE EVALUATION OF THE DIFFERENT CLUSTERING ALGORITHMS FOR THE BOTSWANA IMAGE [NMI, AMI, ARI, FMI, V-MEASURE, COMPLETENESS, HOMOGENEITY, AND ACC (%)]

Evaluation Metrics	K-means	SSC	USPEC	SGCNR	SSSC	S^5C	CAE	VAE	DCN	DEC	S^2DMSC
NMI	62.44	-	67.14	60.06	67.22	54.94	56.35	45.58	62.07	63.95	**75.38**
AMI	62.04	-	66.87	59.62	**66.87**	66.78	55.89	44.99	61.74	63.59	63.59
ARI	42.45	-	49.81	36.45	49.81	51.68	33.06	24.34	39.91	45.76	**59.61**
FMI	48.07	-	54.60	44.74	54.60	41.19	39.07	31.73	46.16	50.78	**63.42**
V-measure	62.44	-	67.14	60.06	67.22	54.94	56.35	45.58	62.07	63.95	**75.38**
Completeness	64.78	-	69.03	65.93	69.27	55.73	58.00	48.01	66.52	66.52	**78.61**
Homogeneity	60.27	-	65.34	55.14	65.29	54.18	54.80	43.39	58.17	61.58	**72.41**
ACC	54.80	-	63.76	46.58	64.44	52.52	46.83	41.26	55.85	60.16	**67.67**

in Table VI, S^2DMSC outperforms other clustering methods and achieves the best clustering performances, which further illustrates the effectiveness of S^2DMSC.

4) Botswana Image: The NASA EO-1 satellite acquired Botswana image over the Okavango Delta, Botswana, from 2001 to 2004. The Botswana image contains 145 spectral bands with a spectral coverage range of 400–2500 nm. The Botswana image consists of 14 different land-cover classes with the size of 1476 × 256, and the spatial resolution is 30 m. Fig. 6 shows the result maps that are obtained by different representative clustering methods. Tables VII and VIII represent the clustering quantitative evaluation of different clustering mechanisms.

From Fig. 6 and Table VIII, we can clearly observe that the S^2DMSC has higher performances on the Botswana image in contrast to the other clustering methods. Among the clustering

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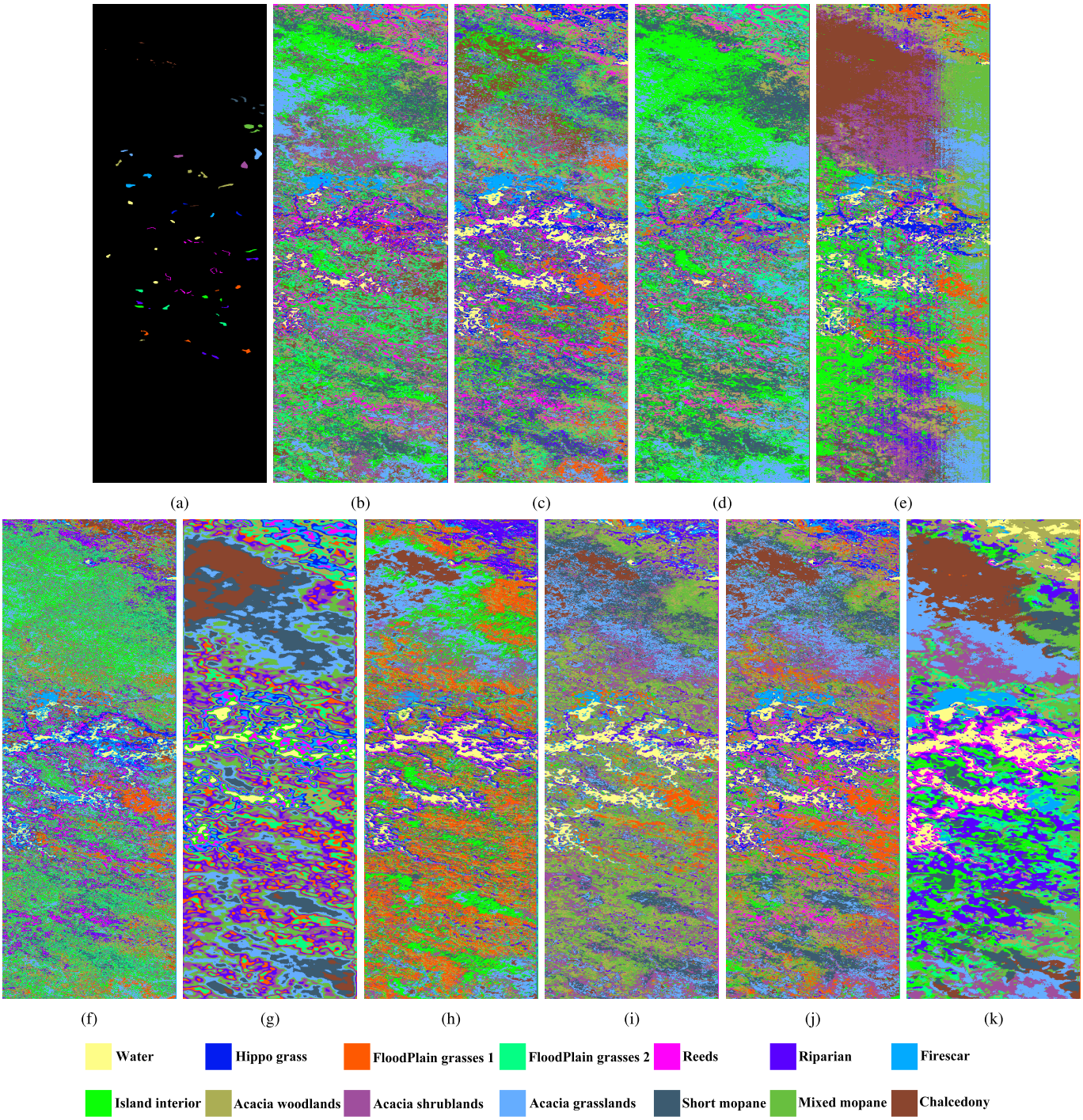

Fig. 6. Clustering maps of Botswana image. (a) Ground truth. (b) K-means. (c) USPEC. (d) SGCNR. (e) SSSC. (f) S^{5}C. (g) CAE. (h) VAE. (i) DCN. (j) DEC. (k) S^{2}DMSC.

performances acquired by the subspace-based clustering methods, the SSSC has much higher clustering results than the USPEC, SGCNR, and S^{5}C, and the clustering accuracy OA improved by 0.68%, 17.86%, and 11.92%, respectively. For the deep learning-based clustering methods, DEC achieves a better clustering map than CAE, VAE, and DCN. Compared with the K-means and DEC, SSSC can significantly improve the clustering accuracy OA and improved by 9.64% and 4.28%, respectively. At the same time, we can also find that among all the comparison clustering methods, the CAE and VAE achieve poor clustering performances, because the Botswana image contains complex background information resulting in many misclassifications. Compared with the K-means, SSSC, and DEC, the proposed S^{2}DMSC performs better and achieves an improvement in terms of the clustering accuracy OA of almost 12.87%, 3.23%, and 7.51%, respectively. Therefore, the S^{2}DMSC significantly outperforms other clustering methods both quantitatively and visually and achieves the best clustering performance with a clustering accuracy OA of 67.67%.

In addition, Table VIII denotes the clustering accuracy evaluation metrics on the Botswana image. Table VIII shows that S^{2}DMSC achieves better evaluation performances than the

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TABLE IX
QUANTITATIVE EVALUATION WITH OR WITHOUT MULTILEVEL FEATURES [OA (%)]

Data sets	Without Multi-Level Features	With Multi-Level Features
Indian Pines	47.21	54.21
Salinas	69.08	72.16
Pavia Center	87.70	90.51
Botwana	65.82	67.67

other clustering methods. It further illustrates the superiority of S^{2}DMSC.

C. Parameter Analysis

The S^{2}DMSC studies five important parameters that have a certain impact on the clustering performance: parameter λ, the number of initial superpixels $W^{(0)}$, GConv operation, convolution kernel size, and network depth. In this article, we utilize the cross-validation method to evaluate the sensitivity of each parameter in detail. The related variation curves of the clustering accuracy OA are given in Supplemental Material for each parameter on the Indian Pines image, Salinas image, Pavia Center image, and Botswana image.

1) Parameter λ: The parameter λ is an important parameter of the S^{2}DMSC, which is utilized to offset the relationship between sparse regularization and reconstruction error. The parameter λ, as an important regularization parameter in the sparse recovery algorithm, adopts the same strategy as [8] and [11], and the parameter λ is described as follows:

$$\lambda = \alpha/u$$
$$u = \min_i \max_{j \neq i} \left| \mathbf{y}_i^T \mathbf{y}_j \right|$$

where α denotes the adjustment coefficient, u is related to the particular dataset, and $\mathbf{y}_i$ and $\mathbf{y}_j$ denote the ith hyperspectral pixel and the jth hyperspectral pixel, respectively. Therefore, the sensitivity of the parameter λ is related to the parameter α and the specific HSI. On the four hyperspectral datasets, the parameter α is set to range from 100 to 1000. Fig. S-2 in Supplemental Materials denotes that the clustering accuracy OA changes with the value of parameter α. As shown in Fig. S-2, it can be illustrated that the quantitative clustering result of parameter α for the four hyperspectral datasets is relatively stable. Specifically, the clustering accuracy OA of the proposed S^{2}DMSC obtains the best performances when the parameter α is set to [500, 700] for all four HSIs.

2) Parameter $W^{(0)}$: The parameter $W^{(0)}$ represents the number of initial superpixels, related to the spatial distribution of HSIs. The parameter $W^{(0)}$ is used as the input superpixels to construct the hierarchical superpixels through the region merging method, which determines the deep multilevel spatial–spectral features. For the four hyperspectral datasets, the parameter $W^{(0)}$ is set to range from 400 to 3000. Fig. S-3 in Supplemental Material denotes that the clustering accuracy OA changes with the value of parameter $W^{(0)}$. As can be clearly seen in Fig. S-3, the number of the initial superpixels is important for S^{2}DMSC. Specifically, the quantatitative clustering result in terms of quantitative evaluation OA achieves superior clustering accuracy for Indian Pines image when the parameter $W^{(0)}$ is set to 1000; the quantitative clustering accuracy OA achieves the superlative performance for Pavia Center dataset and Salinas dataset when the parameter $W^{(0)}$ is set to 2600; the quantitative clustering accuracy OA achieves the superlative performance for Botswana dataset when the parameter $W^{(0)}$ is set to 2200.

3) GConv Operation: The GConv operation is utilized to aggregate the long-range spatial similar information of HSIs. It determines the quality of similarity for superpixel. The range of the number of GConv operations for each layer feature is set from 0 to 5. Fig. S-4 in Supplemental Material denotes that the clustering accuracy OA changes with the number of GConv operations. From Fig. S-4, we can observe that the performance of the proposed S^{2}DMSC is relatively stable, and the best performance of the proposed S^{2}DMSC is obtained when the number of GConv is set to 1 for all four HSIs. It further illustrates that the first-order proximity is very important for the long-range spatial similarity in HSIs.

4) Convolution Kernel Size: The convolution kernel size denotes the size of the receptive field. The range of convolution kernel size for each of the four HSIs is set from 1×1 to 9×9. Fig. S-5 in Supplemental Material denotes that the clustering accuracy OA changes with the size of the receptive field. From Fig. S-5, we can observe that the S^{2}DMSC performs good in most cases and has best performance with the highest OA when the size of the receptive field is set to 5×5 for all four hyperspectral datasets.

5) Network Depth: The network depth is a crucial hyperparameter, which has a significant impact on the clustering performance. In general, a deepening network does not necessarily improve clustering performance, because the gradient disappears. The range of network depth for each of the four HSIs is set from 2 to 5. Fig. S-6 in Supplemental Material denotes that the clustering accuracy OA changes with the depth of CNN. As shown in Fig. S-6, the influence of clustering accuracy is relatively stable for various values of network depth, and good performance with the highest OA is obtained when network depth is set to 3 for all four hyperspectral datasets.

D. Ablation Study

To spotlight the advantages from the multilevel features in the proposed S^{2}DMSC, we employ the ablation study to validate its effectiveness. Specifically, the comparison clustering accuracies of quantitative evaluation OA with and without multilevel features are shown in Table IX.

As shown in Table IX, multilevel features are incorporated into the SSC module to comprehensively combine high-level and low-level semantic information of HSIs. Compared with self-expression features using the final output features of the CNN module, incorporating multilevel features into the self-expression module can better explore the intrinsic deep spatial–spectral features of HSIs and achieve superior clustering performances.

E. Comparison of Running Time

In this section, the average running time of different methods on each dataset is described in Table X. The experimental

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TABLE X
RUNNING TIME (s) OF DIFFERENT METHODS ON EACH DATASET (SECONDS)

Datasets	K-means	SSC	USPEC	SGCNR	SSSC	S^{5}C	CAE	VAE	DCN	DEC	S^{2}DMSC
Indian Pines	8.51	1181.38	2.23	62.32	5.24	307.01	153.39	20.74	625.24	218.85	756.51
Salinas	23.87	-	4.07	91.69	40.46	1343.40	183.18	57.79	3767.58	1046.83	1696.01
Pavia Center	73.14	-	15.10	707.66	122.91	5298.25	840.24	166.67	15301.41	5976.45	6453.79
Botawana	54.80	-	20.24	819.30	734.48	5054.60	778.93	234.69	9206.33	3348.30	5309.90

environment for all methods consists of an i7-8700K CPU and a GTX-3090 GPU.

From Table X, we can observe that the scalable SC methods require a shorter running time in contrast to the SSC in the Indian Pines image, because it reduces the high memory capacity and computational load for sparse recovery-based SSC. In particular, as the number of samples increases, the SSC can cause out-of-memory. For the deep clustering-based methods, the DCN requires more running time. Compared with the SSC, the S^{2}DMSC requires a shorter running time in the Indian Pines image. Moreover, as the number of samples increases, the running time of the proposed S^{2}DMSC slowly decreases compared with DCN, which shows the promise of the S^{2}DMSC for large HSIs.

V. CONCLUSION

In this article, deep multilevel SSC with self-supervised CNN is proposed for simultaneous learning deep multilevel spatial–spectral features and high-quality pseudo-label information, consisting of three modules: CNN module, MSC module, and high-quality pseudo-label-based self-supervision module. Specifically, the CNN module extracts complex nonlinear structural information of HSIs, the MSC module combines high-level and low-level spatial–spectral features to promote clustering performance, and the pseudo-label self-supervision module selects high-quality pseudo-labels generated by the SSC to supervise the learning of CNN and self-expression modules. In addition, the deep multilevel spatial–spectral features extracted from hierarchical superpixels are proposed to solve the limitations of computational load and memory capacity in the SSC for large HSIs. Experiment results on the HSIs have verified the rationality and effectiveness of the proposed S^{2}DMSC.

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